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Editors
Systematic Biology

Dear Editors,

Please consider the Research Article "Strong Tree-Childness Is a Sharp Generic-Identifiability Boundary for Level-2 Jukes–Cantor Networks". The paper asks which reticulate evolutionary histories are recoverable, even in principle, from an exact molecular site-pattern distribution.

The main theorem gives an if-and-only-if classification for binary, already-simple, strongly tree-child level-2 semi-directed networks under the open four-state Jukes–Cantor model. A source-relative full-dimensional containment occurs exactly when the labelled reduced bridge trees agree and corresponding blobs are isomorphic or differ by ordinary triangle redirection. Thus there are no proper one-sided generic containments, and the topology is generically reconstructible modulo that single local move. The boundary is sharp: for every number of taxa at least four, explicit triangle-free weakly-but-not-strongly tree-child pairs have a common regular full-dimensional stochastic region.

The result extends triangle-free level-2 identifiability to the complete strong level-2 class under the convention stated in the manuscript and connects algebraic identifiability directly to what evolutionary network features exact sequence data recover. The finite graph-to-polynomial part is supported by exact primary and separately implemented replay certificates, mutation tests, and a deterministic release archive.

Generative AI systems assisted with exploratory algebra, implementation, drafting, and adversarial review. Earlier automated claims that failed audit were rejected and quarantined. The manuscript contains a full disclosure; I accept responsibility for all claims, code, and submission. No human specialist review is claimed.

I confirm, subject to the final checks recorded in the accompanying human checklist, that this manuscript is original, is not under consideration elsewhere, and has one author. No specific funding supported the work, and I declare no competing interests. The code and exact certificates are linked in the manuscript and submission metadata.

Thank you for your consideration.

Sincerely,

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